FUN-3.B.3 Rearranging tangent, cotangent, secant, and cosecant functions using identities allows differentiation using derivative rules.

2-10 Derivatives of Other Trigonometric Functions

Previously (Topic 2.7), we learned the derivatives for $f(x) = \sin(x)$ and $f(x) = \cos(x)$.

Now, we will include the remaining four trigonometric functions so that we can work with all 6 trig functions.

Derivatives of Trigonometric Functions	
Function	**Derivative**
$f(x) = \sin(x)$	$f'(x) = \cos x$
$f(x) = \cos(x)$	$f'(x) = -\sin x$
$f(x) = \tan(x)$	$f'(x) = \sec^2 x$
$f(x) = \cot(x)$	$f'(x) = -\csc^2 x$
$f(x) = \sec(x)$	$f'(x) = \sec x \tan x$
$f(x) = \csc(x)$	$f'(x) = -\csc x \cot x$

FYI, the derivatives for tangent, cotangent, secant, and cosecant can be found by using trigonometric identities and the quotient rule!

Example 1: Find the derivative of $f(x) = \sin x \tan x$

$$f'(x) = \cos x \tan x + \sin x \cdot \sec^2 x$$

$$= \cancel{\cos x} \cdot \frac{\sin x}{\cancel{\cos x}} + \sin x \sec^2 x$$

$$= \sin x + \sin x \sec^2 x$$

Example 2: Let $y = x^3 \tan(x)$. Find the slope of the line tangent to the graph of y at $x = \frac{\pi}{6}$.

$$y' = 3x^2 \cdot \tan x + x^3 \cdot \sec^2 x$$

$\sec\frac{\pi}{6} = \frac{2}{\sqrt{3}}$

$\left(\frac{2}{\sqrt{3}}\right)^2$

$$y'\left(\frac{\pi}{6}\right) = 3\left(\frac{\pi}{6}\right)^2 \tan\frac{\pi}{6} + \left(\frac{\pi}{6}\right)^3 \sec^2\left(\frac{\pi}{6}\right)$$

$$= 3\left(\frac{\pi^2}{36}\right)\left(\frac{1}{\sqrt{3}}\right) + \frac{\pi^3}{6^3}\left(\frac{4}{3}\right)$$

$$= \frac{3\pi^2}{36\sqrt{3}} + \frac{4\pi^3}{6^3(3)}$$

Example 3: Estimate the derivative with a calculator of $g(x) = \csc^2(4x)$ at $x = 2$.

$= \frac{1}{\sin^2(4x)}$ $\quad Y_1 = 1/(\sin(4x))^2$

$g'(2) \approx 1.202$